

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each



### The Core Components Inside Your PC

- **Motherboard (The Backbone):** A large circuit board that acts as the main hub, physically connecting and allowing communication between every single part of the computer.
- **CPU / Central Processing Unit (The Brain):** A small, powerful chip that handles all the calculations, processes commands, and tells the other components exactly what to do.
- **CPU Cooler / Heatsink (The Thermostat):** A combination of a metal block and a fan sitting directly on top of the CPU to draw away intense heat and keep it from overheating.
- **RAM / Random Access Memory (The Short-Term Memory):** Super-fast, temporary memory sticks that hold the data your active apps (like your open browser tabs) need right this second.
- **GPU / Graphics Card (The Artist):** A dedicated card (or built-in chip) responsible for rendering all the images, videos, and complex 3D graphics you see on your monitor.
- **Storage Drive - SSD or HDD (The Filing Cabinet):** The permanent storage warehouse where your operating system, games, photos, and personal files live even when the PC is turned off.
- **PSU / Power Supply Unit (The Heart):** A metal box usually tucked at the bottom or top that plugs into the wall and converts raw electricity into safe, usable power for all your components.
- **Case Fans (The Exhaust System):** Fans attached to the walls of the cabinet that constantly pull fresh, cool air inside and push suffocating hot air out.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.  
  
**Hint:** Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



### Desktop PC Disassembly Guide



**CRITICAL SAFETY NOTE:** Always turn off the PC, unplug the power cord from the wall, and press the PC's power button once while unplugged to drain any leftover electrical charge before touching any internal parts.

#### Step 1: External Disconnect & Workspace Prep

- **Action:** Unplug all external cables (Power cord, HDMI/DisplayPort monitor cables, mouse, keyboard, and audio jacks). Place the PC tower flat on a clean, non-carpeted table.
- **Caption:** *Clearing all external connections to ensure the tower can be moved and laid flat safely without straining any ports.*

### **Step 2: Removing the Side Panel**

- **Action:** Locate the thumbcrews or latch on the back edge of the main side panel (usually the left side if looking from the front). Unscrew them and gently slide the panel backward to remove it.
- **Caption:** *Removing the main tempered glass or metal side panel to expose the internal components.*

### **Step 3: Disconnecting Power & Data Cables**

- **Action:** Unplug the main power cables attached to the motherboard (the massive 24-pin cable on the right side and the 4/8-pin CPU power cable at the top left). Next, unplug the wide power cables and thin data cables connected to your storage drives.
- **Caption:** *Unplugging the main electrical supply lines and storage cables from the motherboard to prevent tangles.*

### **Step 4: Removing the Graphics Card (GPU)**

- **Action:** Unscrew the GPU from the rear metal bracket of the case. Press down on the small plastic release latch at the very back of the PCIe motherboard slot, then pull the GPU straight up.
- **Caption:** *Releasing the expansion slot lock and unscrewing the graphics card to safely lift it out.*

### **Step 5: Removing the RAM Modules**

- **Action:** Push down on the small plastic clips at the top and bottom ends of the RAM slots. The memory sticks will pop up slightly on their own; lift them straight out by their edges.
- **Caption:** *Opening the retention clips on the memory slots to slide out the RAM sticks.*

### **Step 6: Removing Storage Drives (SSD / HDD)**

- **Action:** For M.2 SSDs, remove the tiny single screw holding the stick flat and slide it out. For 2.5-inch SSDs or traditional HDDs, unscrew them from their mounting bays or slide out their plastic trays.
- **Caption:** *Unscrewing and extracting the permanent storage drives from their designated mounting locations.*

### Step 7: Removing the Motherboard

- **Action:** Locate and remove the 6 to 9 small screws scattered across the face of the motherboard that secure it to the case wall. Gently lift the motherboard up and out of the case by holding onto its heavy metal heatsinks (do not lift by the fragile chips!).
- **Caption:** *Removing the final mounting screws to lift the entire motherboard assembly out of the empty cabinet.*

3. **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**



## Desktop PC Reassembly Guide

### Step 1: Install the CPU, RAM, and M.2 SSD *Outside* the Case

Before screwing the motherboard into the dark, cramped computer case, lay it flat on top of its cardboard box. Install the CPU into its socket, snap the RAM sticks into their slots until they click, and screw down your tiny M.2 SSD.

- **Why the order matters:** Motherboards are wide open when outside the case. Trying to squeeze your fingers inside a dark cabinet to line up a tiny M.2 screw or press down stubborn RAM sticks is incredibly frustrating and increases the risk of your screwdriver slipping and scratching the board.

### Step 2: Mount the Motherboard into the Cabinet

Carefully lift the motherboard and line it up with the small metal screw holes (called standoffs) inside the case. Secure it using the motherboard screws.

- **Why the order matters:** The motherboard must go in before any major cables or large parts, because everything else plugs directly into it.

### Step 3: Mount the Power Supply Unit (PSU) & Route Cables

Slide the power supply brick into its bay (usually at the bottom back of the case) and screw it tight. Take the thick **24-pin motherboard cable** and the **8-pin CPU power cable** and snake them through the case holes to plug them into the motherboard.

- **Why the order matters:** These power cables are thick and stiff. Plugging them in right now—while the center of the case is completely empty—gives your hands plenty of room to wrestle them into place before bulky components get in the way.

### Step 4: Install Traditional Storage Drives (SATA SSDs / HDDs)

Slide your larger 2.5-inch or 3.5-inch drives into their plastic cages or brackets. Connect their flat SATA data cables to the motherboard, and plug in their power delivery cables from the PSU.

- **Why the order matters:** Like the power cables, storage cables require precise routing. Doing this now ensures your cables are neatly managed before the final large components block access to the SATA ports on the edge of the motherboard.

### Step 5: Install the Graphics Card (GPU)

Locate the top long PCIe x16 slot on your motherboard. Push the plastic tab back, line up the gold contacts of your GPU, and push straight down until the slot clicks shut. Screw the metal bracket to the case, and plug in its dedicated PCIe power cables from the power supply.

- **Why the order matters: The GPU should almost always go in last.** Graphics cards are massive bricks of plastic and metal. If you install the GPU early, it acts like a giant wall inside your case, completely blocking your view and preventing your hands from reaching the motherboard's bottom headers, power plugs, and storage slots.

### Step 6: Final Visual Check & Side Panel Close

Double-check that every cable is fully clicked in and that no loose wires are touching any fan blades. Slide the side panel back on and tighten the thumbcrews. Plug your power cord back into the wall, flip the power switch on, and boot up your freshly rebuilt PC!

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

## Part 1: The Tools Checklist

You don't need a massive toolbox to build a PC. In fact, you can do 95% of the work with just one or two basic tools:

-  **Magnetic Phillips Head #2 Screwdriver**
  - *Why it's important:* This is the universal PC building tool. Almost every screw inside a computer—from the motherboard to the power supply—uses a #2 Phillips head. A **magnetic** tip is crucial because it holds onto the tiny screws, preventing them from falling into dark, unreachable crevices inside the case.
-  **Zip Ties or Velcro Straps**

- *Why it's important:* Loose wires look messy, but more importantly, they can get caught in spinning cooling fans and stop them from working. Bundling them together keeps the interior neat and airflow paths clear.
-  **A Good Flashlight or Smartphone Light**
  - *Why it's important:* Computer cases are deep, dark metal boxes. When you are trying to line up tiny front-panel cables or find a screw hole at the bottom of the case, you will need a direct light source to see what you are doing.

## ⚡ Part 2: The Safety Precautions Checklist

Your body naturally builds up static electricity (especially walking on carpets). If that static shocks a sensitive computer chip, it can permanently fry the electronics.

-  **Unplug the Power Cable (And Drain the System)**
  - *Why it's important:* Turning the PC "off" isn't enough; electricity still trickles into the motherboard. Always unplug the main power cord from the wall. Once unplugged, **press the PC's power button 2–3 times**. This drains any residual electricity left hiding inside the power supply's capacitors.
-  **Work on a Hard, Flat Surface (No Carpets!)**
  - *Why it's important:* Set up your workspace on a wooden, plastic, or glass table, and stand on bare tile or hardwood floors. **Never** assemble a PC on a carpet, bed, or rug, as fabrics are massive breeding grounds for static electricity.
-  **Ground Yourself Frequently**
  - *Why it's important:* You need to constantly discharge static from your body. You can do this easily without buying fancy gear: just occasionally touch a bare, unpainted metal part of your computer case chassis before reaching for a sensitive component like the CPU or RAM.
-  **Remove Jewelry and Roll Up Sleeves**
  - *Why it's important:* Metal rings, watches, or bracelets can accidentally scratch a circuit board track or cause an electrical short. Loose long sleeves or dangling hoodies can get snagged on delicate motherboard pins and bend them out of shape.
-  **Handle Components by Their Edges**
  - *Why it's important:* The gold pins and green/black circuits on your RAM, CPU, and motherboard are incredibly sensitive to the natural oils and moisture on human fingers. Always pick up parts by their tough plastic or bare metal outer edges to keep the electrical contacts perfectly clean.